

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	BE-CSE-AI	Date:	

Code of Conduct

*I **S.Muskan(192472182)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **S.Muskan**

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student). **Constraints:**

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. Source Sentence: "The bank by the river flooded after the storm, but it was saved by quick action."

Note: "bank" is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.

2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Q1. Discourse, Reference Resolution & Text Coherence

Given Paragraph

“John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home.”

a) Referring Expressions and Possible Antecedents

The main referring expressions are **He, She, it, him, they all**, and the newly introduced mention **the dog**.

Referring Expression	Possible Antecedents	Initial Interpretation
He	John, Mary, dog	John
She	John, Mary, dog	Mary
it	ball, park	ball
him	John, dog	John
they all	John, Mary, dog	John + Mary + dog

b) Applying the Constraints

The constraints are applied in the following order:

1. **Gender and number agreement**
2. **Semantic compatibility**
3. **Coherence**
4. **Recency as a soft preference**

1. “He”

- John → male singular → compatible.
- Mary → female → eliminated.
- Dog → gender not specified → cannot be selected confidently.

Therefore:

He → John

2. “She”

Possible antecedents: John, Mary, dog.

- John → male → eliminated.
- Mary → female singular → compatible.
- Dog → gender unspecified → less suitable.

Therefore:

She → Mary

3. “it”

Possible antecedents: ball, park.

The phrase “**play with it**” is semantically compatible with a ball but not naturally with the park.

Therefore:

it → ball

4. “him”

Possible antecedents: John, dog.

The immediately preceding mention is **the dog**, so recency initially favors the dog. However, interpreting the sentence as the dog chasing itself is semantically inappropriate.

The dog chased the dog

↓

Self-chasing

↓

Semantically unsuitable

John is a valid animate male antecedent and fits the action “**chased him.**”

Therefore:

him → John

5. “they all”

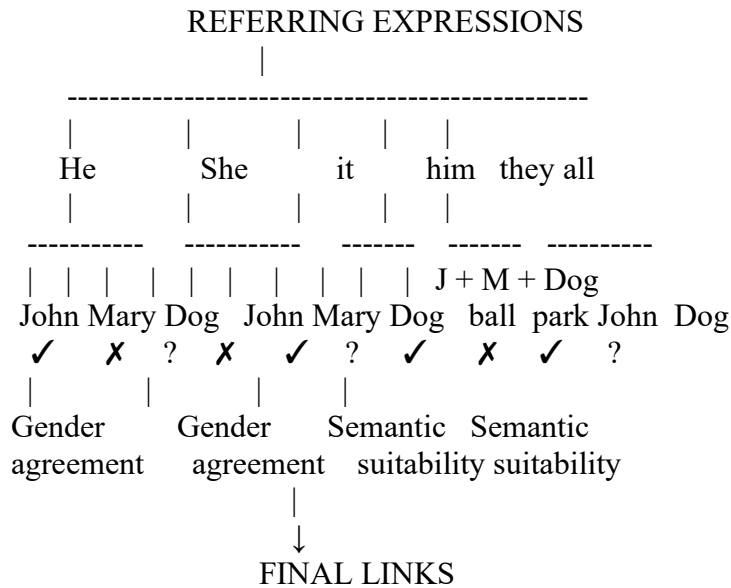
The plural expression refers to the three active discourse entities:

John + Mary + dog

Therefore:

they all → John, Mary and the dog

c) Constraint Graph / Coreference Analysis



Constraint Table

Variable	Candidate	Gender/Number	Semantic Compatibility	Coherence	Result
He	John	✓	✓	✓	Selected
He	Mary	✗	✓	✗	Eliminated
She	John	✗	✓	✗	Eliminated
She	Mary	✓	✓	✓	Selected
it	ball	✓	✓	✓	Selected
it	park	✓	✗	✗	Eliminated
him	John	✓	✓	✓	Selected
him	dog	?	✗	✗	Eliminated
they all	John + Mary + dog	✓ plural	✓	✓	Selected

d) Final Coreference Chains

John —————> He —————> him —————> they

↑

Mary —————> She —————> they

↑

ball —————> it

dog —————> they

Final Resolved Paragraph

“John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.”

This version removes the ambiguous pronouns while preserving the original meaning and entity relationships.

e) Priority of Constraints

The priority is:

Gender/Number → Semantic Compatibility → Coherence → Recency

Gender and number are strong grammatical constraints. Semantic compatibility is next because an antecedent must make sense with the action. Coherence ensures that the same entities remain consistent throughout the paragraph. Recency is treated as a **soft constraint** because the nearest mention is not always the correct antecedent.

If recency were relaxed, the system would rely more heavily on semantic compatibility and discourse coherence. For example, although **the dog** is closer to “him,” semantic compatibility rejects a self-chasing interpretation and selects **John**.

Conclusion: Constraint-based resolution produces a coherent entity chain by combining grammatical, semantic and discourse information rather than selecting only the nearest noun.

Q2. Dialog and Conversational Agents, Discourse Planning & Surface Realization

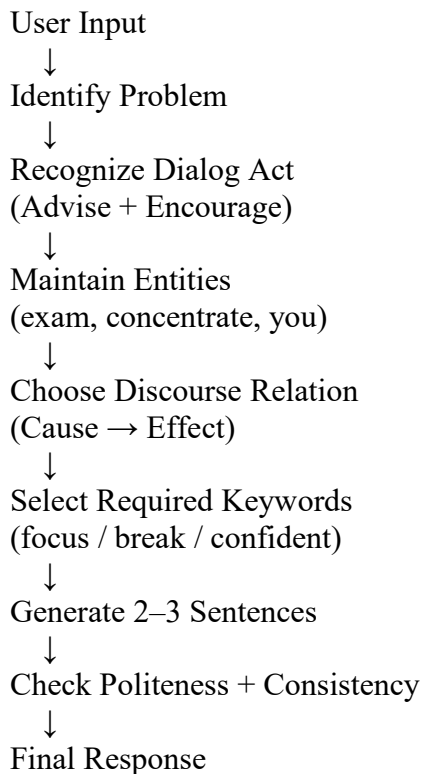
Given Conversation

User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage.

a) Discourse Planning Steps

Before generating the response, the following steps are required:



Constraint Checklist

Constraint	Requirement
Dialog Act	Advise + Encourage
Entity coherence	exam, concentrate, you
Discourse relation	Cause-Effect or Elaboration
Keywords	At least two of focus, break, confident
Length	2–3 sentences
Logical consistency	No contradiction
Tone	Polite and positive

b) Three Valid Responses

Response 1

Since the exam is tomorrow, take a short break and then return to your study with one small task to improve your focus. You can concentrate better step by step, and you should feel confident about the exam.

Constraints satisfied:

- ✓ Advice
- ✓ Encouragement
- ✓ exam / concentrate / you
- ✓ Cause-Effect
- ✓ break / focus / confident
- ✓ 2 sentences
- ✓ Positive and logical

Response 2

You are finding it hard to concentrate because the exam is tomorrow, so take a short break before studying again. Start with one topic at a time to improve your focus and feel confident.

Constraints satisfied:

- ✓ Advice
- ✓ Encouragement
- ✓ exam / concentrate / you
- ✓ Cause-Effect
- ✓ break / focus / confident
- ✓ 2 sentences
- ✓ Positive and logical

Response 3

For the exam tomorrow, put your phone away and study in short sessions so you can focus without feeling overwhelmed. Take a break between sessions, and remember that steady preparation can help you feel confident.

Constraints satisfied:

- ✓ Advice
- ✓ Encouragement
- ✓ exam / concentrate / you
- ✓ Cause-Effect
- ✓ focus / break / confident
- ✓ 2 sentences
- ✓ Positive and logical

c) Evaluation and Best Response

Criterion	Response 1	Response 2	Response 3
Advise + Encourage	✓	✓	✓
Entity coherence	✓	✓	✓
Cause-Effect	✓	✓	✓
Required keywords	✓	✓	✓
2–3 sentences	✓	✓	✓
Logical consistency	✓	✓	✓
Positive tone	✓	✓	✓

Best Response: Response 2

Response 2 is the best because the **cause-effect relation is explicit**:

Exam tomorrow
↓
Difficulty concentrating
↓
Take a short break
↓
Study one topic at a time
↓
Improve focus
↓
Feel confident

It directly addresses the user's problem, provides a practical action, and ends with encouragement. It also satisfies every constraint without adding unnecessary information.

d) Effect of Violating Two Constraints

Violation 1: Entity Coherence

If the response does not maintain entities such as **exam** and **concentrate**, it may become disconnected from the user's actual problem.

Example:

“Take a break and watch something. Everything will be fine.”

This does not clearly address the exam or concentration problem, reducing conversational relevance.

Violation 2: Logical Consistency

A contradictory response could confuse the user.

Example:

“Take a break because you need to study continuously without stopping.”

The advice contains conflicting instructions. This reduces trust and makes the dialog less useful.

Conclusion: Constraint-based dialogue generation ensures that the response remains relevant, coherent, polite, and consistent with the user's intention.

Q3. Machine Translation, Discourse Structure, Predicate Logic & Interpretation

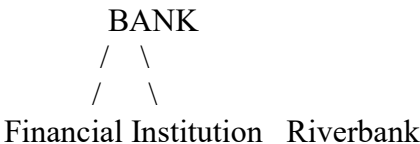
Source Sentence

“The bank by the river flooded after the storm, but it was saved by quick action.”

The assessment identifies “**bank**” as ambiguous between a financial institution and a riverbank.

a) Word Sense Disambiguation

Possible Senses



The phrase:

“**bank by the river**”

provides a strong contextual clue.

A financial bank can be located near a river, but the combination with “**flooded**” strongly supports the physical landform sense. Therefore:

Correct Sense

bank → riverbank

Constraint Analysis

Constraint	Evidence	Result
Context	“by the river”	Supports riverbank
Semantic compatibility	A riverbank can flood	Supports riverbank
Financial sense	A bank may suffer flooding, but “bank by the river” and physical flooding context favor landform	Lower probability
Coherence	Same physical entity is flooded and saved	Riverbank
Information preservation	River and storm relationships retained	Satisfied

Therefore:

BANK = RIVERBANK

Let:

- $b = \text{riverbank}$
- $r = \text{river}$
- $s = \text{storm}$
- $a = \text{quick action}$

Predicates:

- $\text{Bank}(b) = b \text{ is a bank}$
- $\text{River}(r) = r \text{ is a river}$
- $\text{LocatedBy}(b,r) = b \text{ is located by } r$
- $\text{Flooded}(b) = b \text{ flooded}$
- $\text{Storm}(s) = s \text{ is a storm}$
- $\text{After}(\text{Flood}(b), \text{Storm}(s)) = \text{flooding occurred after the storm}$
- $\text{QuickAction}(a) = a \text{ is quick action}$
- $\text{Saved}(b,a) = \text{quick action saved the bank}$

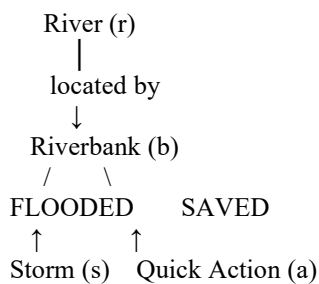
Predicate Representation

$\text{Bank}(b)$
 $\text{River}(r)$
 $\text{LocatedBy}(b,r)$
 $\text{Flooded}(b)$
 $\text{Storm}(s)$
 $\text{After}(\text{Flood}(b), \text{Storm}(s))$
 $\text{QuickAction}(a)$
 $\text{Saved}(b,a)$

Combined representation:

$\text{Bank}(b) \wedge \text{River}(r) \wedge \text{LocatedBy}(b,r)$
 $\wedge \text{Flooded}(b) \wedge \text{Storm}(s)$
 $\wedge \text{After}(\text{Flood}(b), \text{Storm}(s))$
 $\wedge \text{QuickAction}(a) \wedge \text{Saved}(b,a)$

Entity-Relation Representation



This representation preserves the important entities and relations from the original sentence.

c) Coherent Target Sentence / Paraphrase

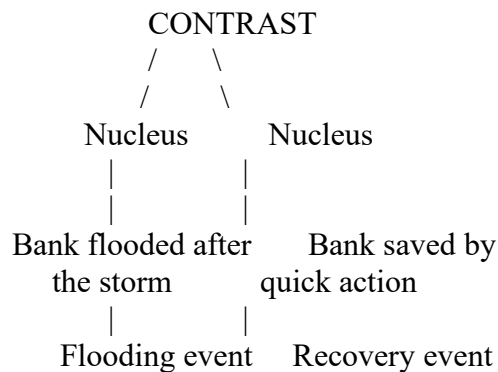
“The riverbank near the river flooded after the storm, but quick action saved it.”

This satisfies all constraints because:

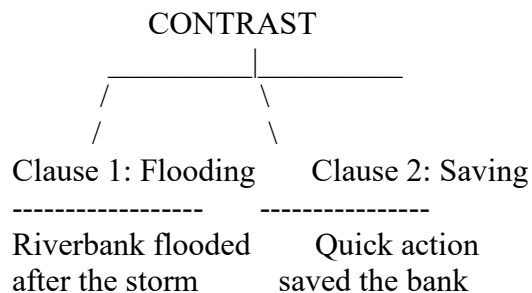
- **Riverbank** resolves the ambiguous word “bank.”
- The **river** is retained.
- The **storm** is retained.
- The flooding event is retained.
- The **quick action** is retained.
- The contrast marker “**but**” is retained.
- The pronoun “**it**” refers coherently to the riverbank.

d) RST-Style Discourse Tree

The word “**but**” establishes a **Contrast** relation between the two clauses.



More explicitly:



The contrast is:

Negative event (flooding) → followed by a positive outcome (saving).

e) Advantages of Constraint-Based Approach over Pure Statistical Machine Translation

A constraint-based approach is preferable for this sentence because it explicitly checks important linguistic requirements before producing the output.

1. Better Word Sense Disambiguation

The context “**by the river**” and “**flooded**” helps select **riverbank** rather than financial institution.

2. Better Information Preservation

The system ensures that the following entities remain present:

Bank/Riverbank
+
River
+
Storm
+
Quick Action

3. Preserves Rhetorical Structure

The conjunction “**but**” represents a **Contrast** relation. A constraint-based system explicitly preserves this relation.

4. Maintains Coherence

The pronoun “**it**” must refer to the same bank/riverbank mentioned earlier.

riverbank → flooded → it → saved
└────────── same entity ─────────┘

5. Reduces Ambiguous Output

A statistical system may select the financial meaning of **bank** based on learned frequency patterns. Constraint-based processing uses explicit contextual and semantic restrictions to prevent this error.

Comparison

Aspect	Constraint-Based Approach	Pure Statistical MT
Word sense	Uses explicit contextual constraints	Depends heavily on learned patterns
Entity preservation	Explicitly checked	May be lost or altered
Discourse relation	Can enforce Contrast	May weaken or change it
Co-reference	Explicitly maintained	Can produce ambiguous references
Interpretability	High	Comparatively lower
Constraint satisfaction	Explicit	Not guaranteed

Final Conclusion

The constraint-based approach gives a more controlled interpretation because it **resolves “bank” as riverbank, preserves the entities and relations, maintains the Contrast relation, and produces a coherent paraphrase**. Therefore, for sentences containing ambiguity and important discourse relationships, explicit constraints provide stronger control over semantic correctness than relying only on statistical prediction.

Overall Conclusion

The three problems demonstrate how constraint-based NLP combines **grammatical agreement, semantic compatibility, discourse coherence, predicate logic, word sense disambiguation, and rhetorical structure**. In Q1, constraints resolve ambiguous references; in Q2, they ensure that a generated response satisfies every dialog requirement; and in Q3, they resolve the word “bank” and preserve the meaning and Contrast relation during paraphrasing.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____